

SEASONAL DATA ANALYSIS

Seasonal Sales Data Analysis – Production Intelligence Project

Description

This project analyzes historical sales data to help production companies identify seasonal demand patterns, improve forecasting accuracy, and optimize production planning. The Jupyter Notebook processes raw sales data, summarizes it into monthly trends, computes yearly averages, and detects true seasonal months—periods where demand consistently rises above normal levels. These insights support inventory planning, manpower allocation, procurement scheduling, and operational decision-making.

Installation

To run this project, install

- Python 3.x
- Jupyter Notebook,
- Pandas, openpyxl, and pywin32.

Project Workflow & Seasonal Logics

1. Importing and Preparing Raw Sales Data

The notebook begins by reading the company's sales dataset and extracting Year, Month, and Month Name. This organizes data into a format aligned with business reporting cycles.

2. Monthly Aggregation

Sales are grouped by Year and Month to calculate Total Sold Quantity. This reveals month-to-month demand patterns and produces a clean dataset for further analysis.

3. Computing Yearly Averages

The notebook calculates the Average Monthly Sales for each year, establishing a performance benchmark. This helps management assess month-level variation against expected demand.

4. Identifying Above-Average Months

Months where sales exceed the yearly average are flagged as potential indicators of high demand. This forms the first layer of seasonal detection.

5. Seasonal Logic – Detecting True Seasonal Demand

A month is labeled seasonal only if it crosses the yearly average AND repeats as a high-performing month in multiple years. This identifies true recurring demand patterns and avoids false seasonal signals.

6. Seasonal Sales Value

For each seasonal month, the exact sales quantity is stored. This data helps estimate required production volume for upcoming peaks and guides resource planning.

7. Month-Year Label Creation

A clear Month-Year label (e.g., "Jan 2023") is created for easy visualization and charting in Excel or dashboards.

8. Exporting Results for Reporting

The notebook produces a clean Excel output showing Month, Year, Total Sales, Averages, Seasonal Flags, and Seasonal Sales Values. This becomes the foundation for presentations, charts, and management reporting.

Business Learnings & Insights

- Identifies peak-demand months
- Enables accurate production forecasting
- Prevents stockouts through proactive planning
- Supports raw material procurement alignment
- Improves manpower, shift, and machine scheduling
- Converts raw data into actionable operational intelligence

Conclusion

The notebook provides a powerful analytical framework for converting historical sales data into clear, strategic business insights. It equips production managers with the information needed to anticipate seasonal demand, optimize production cycles, and strengthen supply chain stability.